Business Statistics

Value-based measures of dispersion



Some graphics and datasets used in this presentation and accompanying MS Excel models are simulated by artificial intelligence. While they are realistic, they are not from actual companies or pre-existing data sets.

Measures of Dispersion Overview

- The dispersion of a variable has two main characteristics:
  - Spread - the distance observations are from the center
  - Shape - how observations are distributed around the center.
- Value-based dispersions
  - Mean Absolute Deviation (MAD),
  - Mean Absolute Percentage Deviation (MAPD)
  - Variance
  - Standard Deviation
  - Coefficient of Variation
  - Skew
  - Kurtosis

Dispersion of a variable

Value-based measures of dispersion

Dispersion of a Variable
A variable takes on multiple values

- As observations have different values, measures of **dispersion** (spread) describe how much those values differ from the center
- Greater dispersion indicates greater variability among the observations
- The observations' dispersion affects:
  - How likely future observations are to be close to the center
  - The variability of calculations based on those observations
- Different measures give different information, so we consider multiple measures to get the whole picture

Value-based measures of dispersion
Calculating the deviation of all observations

- Recall the mean is the value-based measure of "center" for a variable
- Value-based measures depend on how each observation deviates from the mean
- An observation's "deviation" is the observation's value minus the mean
  - For values greater than the mean this results in a positive number
  - For values less than the mean this results in a negative number

$$x_i - \overline{x}$$

- Summing all the deviations **MUST** always = 0
  - The mean was calculated to balance out all the observations
  - Therefore, summing all the deviations cancel out the positive and negative values
- Therefore, to measure the total amount of deviation (dispersion) we must prevent this "cancelling out" effect

Value-based measures of dispersion
Different measures high-level

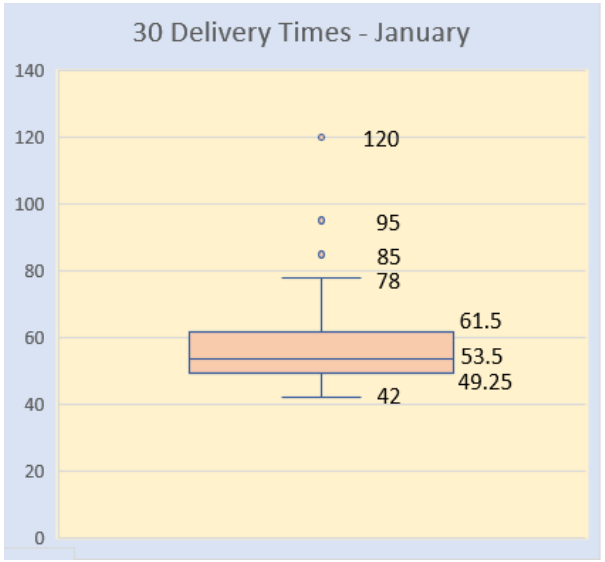
- The measures we will consider are:
  - Mean Absolute Deviation (MAD)
  - Mean Absolute Percentage Deviation (MAPD)
  - Variance
  - Standard Deviation
  - Coefficient of Variation
  - Skew
  - Kurtosis
- MAD and variance use different approaches to prevent positive and negative deviations from canceling each other out

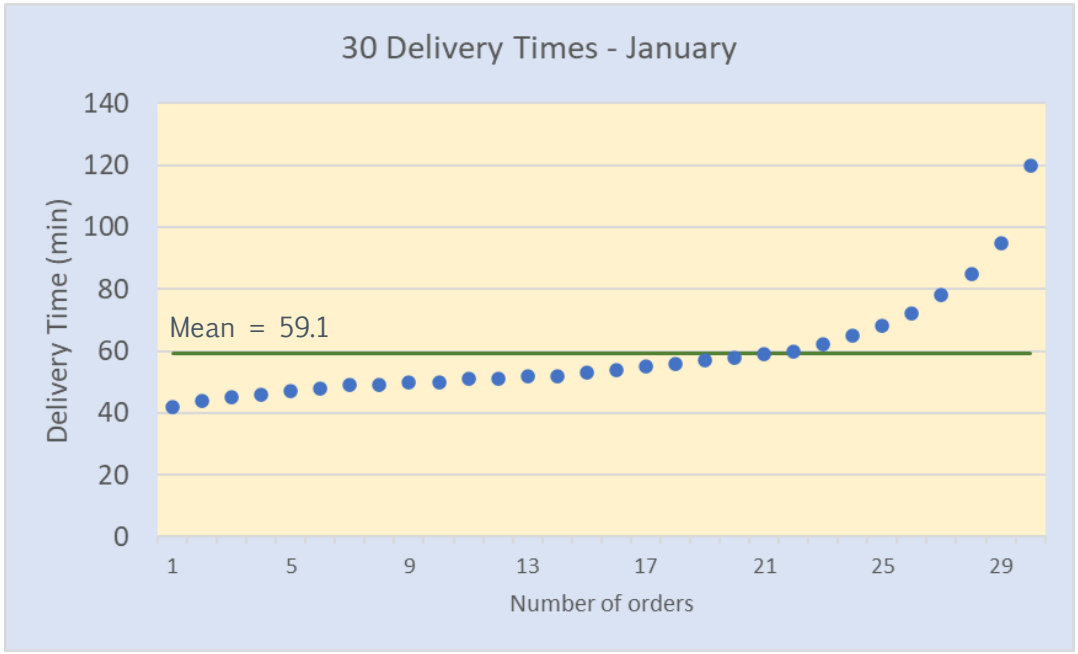
Value-based measures of dispersion
Data set and ordered measures

- An online retailer tracks the delivery time (in minutes) for 30 recent customer orders shipped from the same warehouse in January.
- Management wants to understand typical delivery time, variability, consistency, and whether extreme delays exist.

Delivery Time
46
65
48
55
120
…
50
62
49
42
51

5 Number Summary January				
Min	Q1	Q2	Q3	Max
42	49.25	53.5	61.5	120





Dispersion of a variable

MAD and MAPD

Value-based measures of dispersion
MAD: Mean Absolute Deviation

- MAD:
  - Average of the absolute deviations from the mean for all of the observations
- Deviation:
  - How far an individual observation “deviates” from the mean
  - Shows how spread-out observations are from the “center”
  - Observations greater than the mean are positive, and less than the mean are negative
- Absolute:
  - Summing all of the negative and positive deviations cancel each other out
  - To see the magnitude of all deviations requires preventing this “cancelling out” effect
  - The absolute value returns the positive value for a deviation even if it is negative
- Mean:
  - Sum of the absolute deviations divided by the count of observations

Value-based measures of dispersion
MAD calculations

- Deviation:
  - How far the sample observation x_i "deviates" from the sample mean $\overline{x}$
- Absolute:
  - the absolute value of the deviation $|x_i - \overline{x}|$
- Mean:
  - Sum of the absolute deviations $\sum_i |x_i - \overline{x}|$
  - Divided by the count of observations n

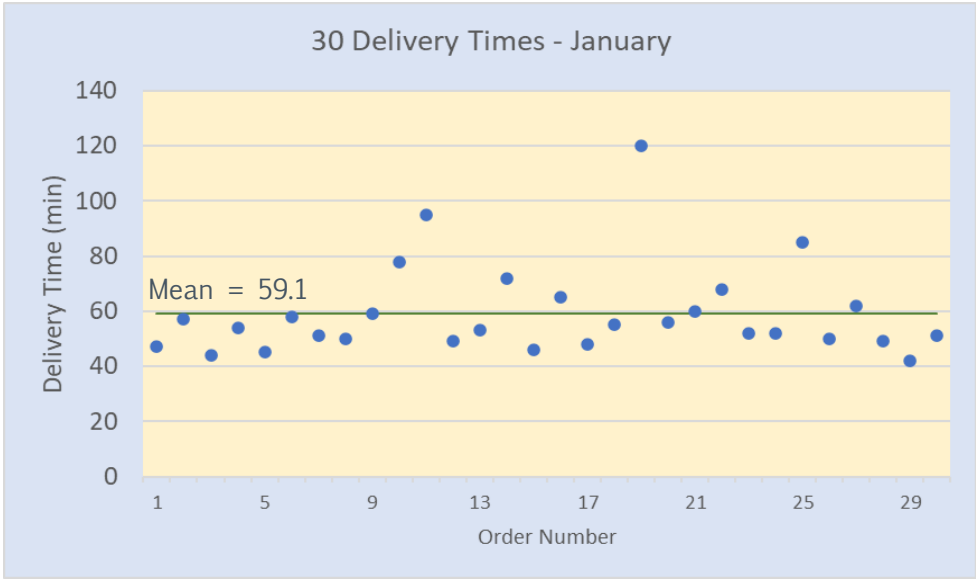
Sample MAD = $\dfrac{\sum_i |x_i - \overline{x}|}{n}$ Excel: =AVEDEV(<data range>)

Population MAD = $\dfrac{\sum_i |x_i - \mu|}{N}$ Excel: =AVEDEV(<data range>)

Value-based measures of dispersion
MAD: delivery time in minutes (mins)

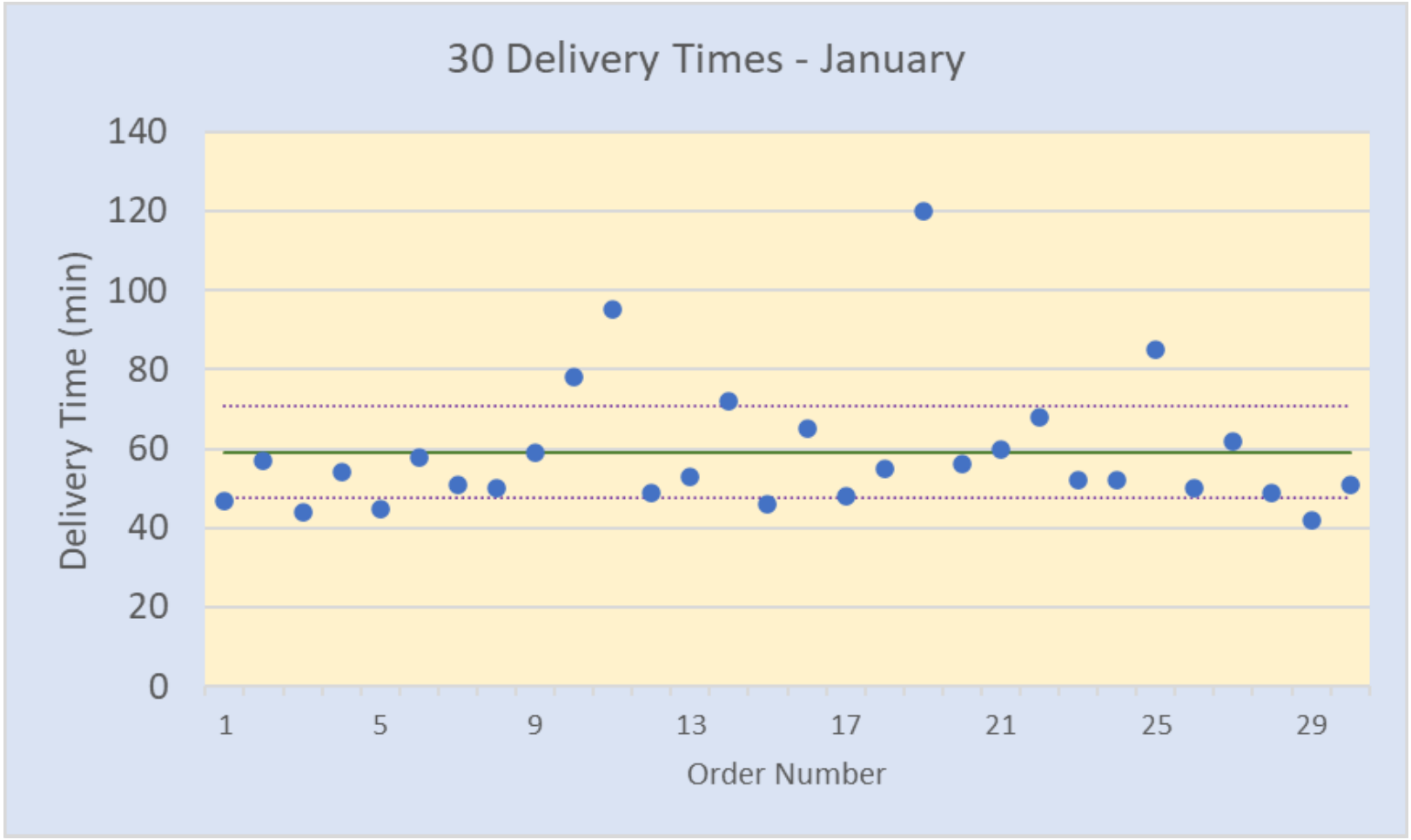
- Deviation:
  - How far observation x_i "deviates" from the mean $\overline{x} = 59.1\ mins$
- Absolute:
  - the absolute value of the deviation $|x_i - \overline{x}| = |x_i - 59.1|$
- Mean:
  - Sum of the absolute deviations $\sum_i |x_i - 59.1|\ mins$
  - Divided by $n = 30$ observations

$$\text{Sample MAD} = \frac{\sum_i |x_i - 59.1|}{30} = \boxed{11.5\ mins}$$



Value-based measures of dispersion
MAD: delivery time in minutes (mins)

- Sample mean: $\bar{x} = \boxed{59.1\ mins}$
- Sample MAD $= \dfrac{\sum_i |x_i - \bar{x}|}{n} = \boxed{11.5\ \text{mins}}$



Value-based measures of dispersion
Is a calculated MAD value good or bad?

- MAD provides an absolute measure of deviation – in this case 11.5 mins
- Does an average deviation of 11.5 mins suggest low or high variability?
- Depends:
  - If the observations range from 1 to 10,000, 11.5 seems extremely low
  - If the observations range from 1 to 30, 11.5 seems horribly high
- This is very important to know especially when comparing two variables
- There needs to be a relative measure to answer this question
- The relative measure for the mean absolute deviation (MAD) is the mean absolute percentage deviation (MAPD)

Value-based measures of dispersion
MAPD: Mean Absolute Percentage Deviations

- The MAD is calculated by dividing the sum of all absolute deviations by **the number of observations**

$$\frac{\sum_i |x_i - \bar{x}|}{n}$$

- The MAPD is calculated by dividing the sum of all absolute deviations by **the sum total of all observations**

$$\frac{\sum_i |x_i - \bar{x}|}{\sum_i x_i}$$

Sample MAPD = $\frac{\sum_i |x_i - \bar{x}|}{\sum_i x_i}$

Population MAPD = $\frac{\sum_i |x_i - \mu|}{\sum_i x_i}$

Value-based measures of dispersion
Mean, MAD, MAPD delivery times (mins)

Sample mean:

$$\bar{x} = \frac{\sum_i x_i}{n} = \frac{\sum_i x_i}{30} = \boxed{59.1\ mins}$$

Sample MAD:

$$MAD = \frac{\sum_i |x_i - \bar{x}|}{n} = \frac{\sum_i |x_i - 59.1|}{30} = \boxed{11.5\ mins}$$

Sample MAPD:

$$MAPD = \frac{\sum_i |x_i - \bar{x}|}{\sum_i x_i} = \frac{\sum_i |x_i - 59.1|}{1{,}773} = \boxed{19.5\%}$$

